

Vessel and segment specific tortuosity metrics

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Named vessels and segments reshape tortuosity

The previously proposed five-stage tortuosity method (local Frenet-Serret descriptors, a Richardson-Mandelbrot scale sweep, retinal-style WAL/WAC whole-tree aggregation plus a Topological Morphology Descriptor, GAMLSS/Mahalanobis reference ranges, and a hand-crafted-features-first ML framing) currently treats “coronary tortuosity” as if it were one well-defined quantity computed the same way regardless of which vessel or which point along a vessel produced it. Both assumptions are wrong, though to different degrees and for different reasons. On vessel identity, the evidence is concrete and, in one case, primary-source-verified: LAD, LCx, and RCA rank oppositely depending on whether tortuosity is scored as a continuous arc-chord ratio or as a discrete bend count, which means the method’s whole-tree aggregation stage (stage 3) and its reference-range stage (stage 4) must run separately per named vessel, not pooled, while the local per-point descriptor stage (stage 1) already is vessel-specific by construction and needs no change. On within-vessel position, the evidence for a systematic proximal-to-distal tortuosity gradient is real in one directly-verified coronary machine-learning paper but otherwise sits at search-summary confidence with no traceable primary source, so this report recommends adding a continuous, arc-length-normalized positional profile alongside the existing whole-vessel scalar rather than either ignoring position or naively averaging landmark-defined segments, the latter being directly undermined by a verified retinal-imaging finding that no tortuosity index satisfies both of the field’s two competing compositionality criteria at once. On metrics, this report locks in ten additions to the method’s descriptor battery, five carried over verbatim from earlier conversation turns and five newly surfaced in this research pass, while explicitly retiring half a dozen candidates that turn out to be the same ratio wearing a different field’s name.

RCA, LAD, and LCx rank tortuosity in opposite orders

The single most load-bearing finding for this section comes from a cohort study comparing non-obstructive and obstructive coronary artery disease patients, which reports median arc-chord Tortuosity Index (TI) by vessel of LCx 1.30, LAD 1.26, and RCA 2.00 in the non-obstructive group, versus a discrete-bend definition (three or more consecutive bends of at least 45 degrees, present in both systole and diastole) that finds “tortuous” prevalence of LCx 40.0 percent, LAD 32.9 percent, and RCA only 9.4 percent in the same patients ([Coronary Tortuosity Index vs. Angle Measurement Method, PMC10795752](#)). The underlying research notes cite this paper with specific medians, interquartile ranges, and p-values but do not carry the explicit full-text-verification tag used elsewhere in the same note file, so it should be read as a directly cited, quantitatively specific source rather than as independently re-verified in this pass. Read at face value, it says something the previously proposed method’s stage 3 (whole-tree WAL/WAC aggregation) cannot currently represent: the

RCA has the *highest* continuous curvature yet the *lowest* discrete-bend prevalence of the three major vessels, the exact opposite ranking under two commonly used tortuosity definitions in the same patients. The paper’s own explanation is that the RCA’s baseline “C-shaped” course around the atrioventricular groove already carries substantial smooth curvature, which continuous arc-chord scoring picks up but which does not resolve into the sharp, discrete 45-degree bends the angle-count definition is looking for. A second, larger cohort of 1,010 patients corroborates the direction of the discrete-bend asymmetry without offering a mechanism: LCx tortuosity prevalence of 26.9 percent, LAD 21.1 percent, RCA only 1.4 percent ([Clinical Implication of Coronary Tortuosity in Patients with Coronary Artery Disease, PLOS ONE](#)), and a tortuosity-severity-index study finds roughly 80 percent of all detected tortuosity concentrated in the LAD, mostly at two specific SYNTAX-score segments, after gating analysis to vessels 2.5 mm or wider in caliber ([Validity of tortuosity severity index, PMC6303525](#)). One further claim, that the RCA displaces roughly twice as much as the left coronary system during the cardiac cycle, is explicitly flagged in the underlying notes as sourced only from a search-engine summary of a PubMed abstract that could not be fetched past a CAPTCHA wall in this research pass; it is mechanistically plausible as a contributor to the RCA’s distinctive tortuosity signature, but it is an open gap, not a verified fact, and should not be quoted as settled.

A structurally identical warning comes from a field with far more statistical power behind it. The UK Biobank retinal-vessel-tortuosity genome-wide association study ran separate artery-specific and vein-specific analyses in addition to a combined one, and the vessel classes turned out to differ not just in mean tortuosity but in the underlying genetic architecture generating it: heritability was $h^2_{SNP} = 0.23$ (SE 0.020) for arteries versus $h^2_{SNP} = 0.15$ (SE 0.021) for veins, with 116 significant loci found for arteries and 63 for veins, and the genes implicated in arterial tortuosity specifically overexpressed in arteries and heart muscle rather than in venous tissue ([Genome-wide Association Studies of Retinal Vessel Tortuosity, PMC10149284](#), verified by direct fetch of the full text). This is a stronger form of evidence than a simple mean-difference finding: it shows that “vessel identity” changes the statistical process that generates the tortuosity value, not merely its average, which is exactly the condition under which pooling across vessel classes produces a number that does not describe any single underlying biological quantity. A third field supplies the clearest quantitative demonstration that this dilution is not merely theoretical. A 2024 study of intracranial arterial tortuosity found a Global Tortuosity Index, standardized mean curvature computed across the entire arterial tree, independently associated with atherosclerotic burden at an adjusted odds ratio of 1.33 (95 percent CI 1.09 to 1.62, $n = 516$), but the same underlying tortuosity construct restricted to the basal arteries alone strengthened that association to an adjusted odds ratio of 1.48 (95 percent CI 1.22 to 1.81) in the identical cohort ([Global intracranial arterial tortuosity is associated with intracranial atherosclerotic burden, PMC11636879](#)). This figure is drawn from an abstract-level search summary in the underlying notes, since the full text was blocked by an authentication redirect and could not be independently confirmed beyond the headline odds ratios, so it should be treated as search-summary-confidence evidence of attenuation rather than a primary-verified case study; it does not rise to a documented Simpson’s-paradox sign reversal, and no field surveyed in this research pass produced one. What it does show, at whatever confidence level the source supports, is a real, same-cohort, same-endpoint example of a pooled tortuosity score performing measurably worse than the identical construct restricted to an anatomically coherent vessel subset, which is the general statistical mechanism Simpson’s-paradox literature attributes to pooling heterogeneous subgroups whose individual associations differ in strength or direction ([Simpson’s Paradox, Wikipedia](#)).

Two evidence gaps deserve stating plainly rather than papering over. No source found in this

research pass gives a numeric tortuosity value for diagonal branches or the left main trunk as distinct from LAD, LCx, and RCA; the coronary literature treats tortuosity as a three-vessel construct, and this report does not assume diagonals or the left main behave like their parent vessel, since that assumption has no evidence behind it either way. And no source decomposes how much of the RCA-versus-left-system tortuosity difference is attributable separately to caliber, length, cardiac-cycle motion, or anatomical course; the mechanism is plausible but not isolated by any multivariable study surfaced here.

Put together, this evidence licenses a concrete, stage-by-stage verdict for the previously proposed method rather than a blanket “stratify everything” instruction, because not every stage of the pipeline pools across vessels in the same way or with the same risk.

Stage	Pools across named vessels today	Verdict
1: local descriptors (κ_a , τ_a , Menger check)	No, already computed per point on each vessel’s own centerline	Keep as is
2: Richardson-Mandelbrot scale sweep (D_κ)	No, fit per vessel	Keep per vessel; do not average the exponent across LAD/LCx/RCA before reporting, since their baseline shape classes differ (RCA’s smooth C-shaped course versus LAD/LCx’s sharper discrete bends)
3: whole-tree aggregation (WAL/WAC)	Yes, by design, into one scalar per tree	Change required. Compute WAL/WAC per named vessel first (LAD, LCx, RCA, reported separately) and treat the fully pooled whole-tree scalar as a secondary, explicitly labeled aggregate, never the primary reported number, given the direct rank-reversal evidence above
3: TMD topological vector	Yes, inherently, since branching topology is a whole-tree property	Stays pooled at the tree level; this is not the kind of pooling the evidence above warns against, because there is no meaningful “per-vessel” topology to separate it into. A per-subtree Horton-Strahler ratio (Section 3) is the appropriate vessel-level companion here, not a replacement

Stage	Pools across named vessels today	Verdict
4: GAMLSS scalar reference ranges	Should be stratified	Change required , matching this project’s own already-locked practice elsewhere of LAD/LCx/RCA-specific reference centiles and a left-main length exception; this section’s evidence is the direct justification for that existing practice, not a new idea
4: MCD/Mahalanobis vector reference range	No, one distance per patient	Stays pooled at the tree level, since the TMD vector it scores is itself a whole-tree object
5: downstream feature vector	Concatenates vessel-specific and whole-tree features	Feed κ_a , TI , and D_κ as separate named features per vessel (mirroring the retinal GWAS’s parallel artery/vein analyses) rather than one pooled scalar, alongside the pooled TMD vector; this is feature-level pooling, a different and lower-risk operation than pooling raw measurements before computing a statistic

Proximal-to-distal differences are real but poorly measured

The clinical segmentation schemes already in wide use, the SCCT 18-segment model and the AHA 15/16-segment model, both divide the LAD and RCA into proximal, mid, and distal thirds and the LCx into proximal and mid-distal, which is a directly reusable precedent if the method adds within-vessel granularity: it would not need to invent new segment boundaries, and doing so would keep results comparable to the CAD-RADS and SCCT reporting categories clinicians already use ([SCCT 18-segment model, ResearchGate](#); [AHA segment model, PMC7860735](#)). Both schemes place segment boundaries at anatomical landmarks, specific branch origins and bends, rather than at fixed millimeter cutoffs, but this comes with a caveat the underlying notes flag explicitly: which landmark governs the proximal/mid LAD boundary is not fully standardized, with some sources drawing the line at the first septal perforator and others at the first diagonal branch. This entire paragraph rests on aggregated web-search summaries of SCCT and AHA guideline documents that returned 403 or CAPTCHA blocks on direct fetch in the underlying research, so the exact landmark wording should be treated as summary-level, not primary-verified, and any implementation that adopts one convention over the other needs to record which one it chose as an explicit, stated deviation, consistent with this project’s own convention of recording departures from published settings.

Whether tortuosity itself actually increases from proximal to distal, independent of disease, is the weakest-evidenced claim this report can respond to, and it should be stated as an open gap rather than a settled finding. The underlying notes report only unattributed, aggregated search summaries for the two most directly relevant claims, that “tortuosity tends to increase” moving distally while calcium volume decreases, and that this effect is attributed to branch-point redundancy producing many small loops distally rather than one broad proximal sweep; neither claim could be traced to a named, fetchable paper in this research pass. No study was found that isolates segment position from disease burden in a matched, angiographically normal cohort. This report does not treat the proximal-to-distal gradient as established fact, and any thesis text drawing on it should say so plainly. What is directly verified, from a paper the underlying research fetched and quoted in full, is a related but distinct finding: a 2026 machine-learning study that fed a transformer model an ordered vector of point-wise local tortuosity along the coronary centerline, rather than a single whole-vessel mean, discriminated coronary artery disease better (AUC 0.67, 95 percent CI 0.65 to 0.69) than a model using only the whole-vessel mean tortuosity plus age and sex (AUC 0.60, 95 percent CI 0.58 to 0.63) ([A Machine Learning Driven Approach to Quantifying Coronary Artery Tortuosity, PMC13308244](#), quoted directly from the fetched article text). This is not a clean, disease-independent proximal/mid/distal comparison, but it is primary-source-verified evidence that *where* along a vessel high tortuosity concentrates carries information a single whole-vessel scalar discards, which is the concrete, verified reason to add positional granularity even without a settled answer on the underlying anatomical mechanism.

The sharper, more actionable finding concerns not whether position matters but what happens once a method tries to represent it. A directly fetched and quoted retinal-imaging paper frames within-vessel partitioning as a compositionality problem and tests two competing mathematical criteria for how a whole-vessel score should relate to its sub-segment scores: Hart’s criterion requires the whole-vessel tortuosity $\tau(V)$ to lie between the tortuosity values $\tau(V_1)$ and $\tau(V_2)$ of any two-way split, while Grisan’s criterion requires the opposite, that $\tau(V)$ exceed every constituent sub-segment’s value, reflecting accumulated curvature across multiple bends. The paper states plainly that these two criteria “cannot be fulfilled simultaneously,” demonstrates on a synthetic three-semicircle test vessel that no tested index satisfies both consistently, and reports that “an increase in the number of sections generally leads to lower global tortuosity values for most combinations” of index and weighting scheme ([Bridging local and global tortuosity of retinal vessels, PLOS ONE](#), directly fetched and quoted). This sharpens a point the earlier “Cross domain” report already flagged but could not fully resolve: that report noted the Hart inequality is “satisfied by construction for any convex combination” and proposed checking it directly on this project’s own coronary centerlines before trusting the WAL/WAC aggregate. The compositionality finding here shows that passing a convex-combination Hart check is not sufficient assurance for a segment-averaging scheme specifically, because how many pieces a vessel is cut into changes the resulting number in a documented direction, and no index satisfies the stricter Grisan criterion at the same time. Concretely: averaging or summing proximal/mid/distal segment scores into a “corrected” whole-vessel number is not a more granular version of the same ground truth, it is a different quantity whose value depends on the partitioning choice itself.

This report’s recommendation follows directly from that finding and from the two structurally different design options the underlying notes identify. Discrete anatomical partitioning at bifurcation, crossing, or endpoint landmarks (the retinal-imaging approach, structurally identical to SCCT/AHA proximal/mid/distal segmentation) requires reliable branch-point detection, which is itself an open sub-problem this project’s own topology-extraction work has not yet solved end to end. A continuous, per-point positional profile requires only a clean ordered centerline and

no branch labeling at all. Given that constraint, this report recommends adding a continuous arc-length-normalized profile of the existing local descriptor, $\kappa_a(s)$ evaluated at the already-locked chord length across normalized arc-length position $s/L \in [0, 1]$, as a companion to the existing whole-vessel scalar, rather than adopting landmark-based proximal/mid/distal segment averaging as the primary representation. This keeps the pipeline’s existing per-point machinery from stage 1 unchanged and adds a positional axis on top of it rather than a new aggregation step that would inherit the same partition-count sensitivity the retinal paper documents. Reporting clinically named SCCT segment labels as metadata alongside the continuous profile, wherever branch-point detection is reliable enough to place them, is worth doing for comparability to clinical reporting categories, but only as an annotation layer on the continuous profile, never as a replacement scoring unit computed by averaging within each labeled bin. Arc-length reparametrization, remapping each vessel’s centerline to a common $[0, 1]$ domain so profiles of different absolute length become directly comparable across patients, is the specific mechanism from functional-data-analysis literature that makes this comparison meaningful, though the underlying notes source this only at the abstract/methods-summary level and it has, as far as this research pass found, never been applied specifically to a coronary vessel ([Arc-Length Re-Parametrization and Signal Registration, Frontiers in Bioengineering and Biotechnology](#)).

One named caveat has to travel with this recommendation rather than sit as a footnote. No source found in this research pass gives a quantified bias curve for tortuosity as a function of proximity to a bifurcation, and a direct search for exactly this artifact returned nothing more specific than an aggregated summary stating the topic was “not explicitly detailed” in the sources it surveyed. The only indirect evidence is that adjacent coronary bifurcation geometry work imposes an explicit minimum standoff, one CFD study reportedly places curvature measurement points no closer than 5 mm to a bifurcation, before trusting a curvature value near a branch point, though this specific figure is sourced only from an aggregated search summary and was not independently confirmed against a primary paper. This report treats a positional profile’s behavior near a bifurcation as a genuine, currently unquantified risk, not a solved problem: any sliding-window or per-point curvature profile implemented for this pipeline should exclude, or flag separately, points within a few millimeters of a detected bifurcation, mirroring the adjacent literature’s standoff convention as a precaution, and should report results as sensitive to that choice rather than presenting one profile as ground truth, exactly as the retinal compositionality paper’s broader point that partitioning choices are never neutral would predict.

Ten simple metrics round out the tortuosity battery

Five of the additions below were already proposed earlier in this conversation, in response to informal follow-up questions, not found fresh in this research pass; they are stated here with their exact formulas and folded into the same final list as the genuinely new items this pass surfaced, per the coordinator’s instruction, rather than re-derived as if newly discovered.

The Inflection Count Metric counts the number of curvature sign changes along a vessel, adds one, and divides by end-to-end distance after multiplying by path length:

$$ICM = \frac{(n_{\text{inflections}} + 1) \cdot L}{D},$$

a formula the underlying research confirmed directly from a cited patent filing describing exactly this construction ([USPTO 8090164](#)). Peak or 95th-percentile local curvature, $\kappa_{\text{peak}} = \max_i \kappa_i(\ell)$ or the 95th percentile of the per-point curvature sequence already produced by stage 1, is an order-statistic

complement to the mean, κ_a , that captures whether a vessel’s total curvature comes from one sharp bend or many mild ones, information the mean alone cannot distinguish. The Horton-Strahler bifurcation ratio, a classical drainage-network descriptor (Horton 1945, Strahler 1957) that orders branch segments recursively (a segment with no tributary is order 1; two merging segments of equal order i produce a downstream segment of order $i + 1$, otherwise the downstream order is the larger of the two) and defines $R_B \approx N_i/N_{i+1}$, the ratio of segment counts between consecutive orders, is proposed as a simpler, single-number companion to the Topological Morphology Descriptor for the branching-topology channel specifically, computable per subtree rooted at each major named vessel without needing the full persistence-diagram machinery. Sinuosity-index classification banding applies fixed thresholds, approximately 1.05, 1.25, and 1.5, to the continuous arc-chord ratio DM or TI already computed, giving an interpretable categorical layer (straight, mildly tortuous, tortuous, severely tortuous) on top of the continuous score, paralleling the kind of banding seen in river-geomorphology sinuosity classification and in vascular-access severity scoring, though with this project’s own chosen cut points rather than either of those fields’ exact thresholds. Raw chord length D and raw arc length L , currently folded only into the ratio $TI = L/D - 1$, are proposed as standalone reported columns in their own right, specifically so the reference-range stage can condition on vessel length as a measured confound rather than leaving it implicit inside a ratio.

The new items this research pass surfaced sit in three groups. The first is a pair of genuinely distinct Bullitt-lineage descriptors, verified directly from a fetched article’s own table, though that article is itself a secondary or tertiary restatement of the original Bullitt methodology rather than the original paper. Bullitt’s Triangular Index, here labeled TI_{local} to avoid a naming collision with this project’s own $TI = L/D - 1$, is a local sliding-triangle straightness average rather than a global ratio:

$$TI_{\text{local}} = \frac{1}{n} \sum_{i=1}^n \frac{a_i + b_i}{c_i},$$

where at each of n sampled points a triangle is built with the apex at that point and two base vertices at a fixed offset ahead and behind on the centerline, a_i and b_i are the two sides from the apex, and c_i is the triangle’s base ([Assessing ML Models for Predicting Age with Intracranial Vessel Tortuosity, PMC10669197](#)). This is genuinely different in kind from the project’s global TI , since it is local and smoothed over a sliding window rather than end-to-end, and it is recommended as an addition on those grounds. Relative Length, $RL = l/l_c$ (chord over arc), and Product-of-Angle-Distance, $PAD = SOAM \times RL$, come from the same fetched table but are not recommended: RL is algebraically the reciprocal of $1 + TI$, the same information already carried by the project’s own secondary descriptor under a different name and direction, and PAD is a multiplicative composite of two other metrics (one of which, $SOAM$, is not itself computed by this project in that exact normalization) rather than an independently informative quantity.

The second group is a well-established clinical precedent and a companion classification scheme from vascular fields entirely outside coronary or cerebral imaging. The Umbilical Coiling Index is computed postnatally as coils per unit cord length, $UCI = (\text{number of complete umbilical vascular coils})/(\text{cord length})$ with a normal population value of 0.17 coils per cm and hyper- or hypocoiled defined by the 90th and 10th percentiles of that distribution ([The umbilical coiling index, PubMed 16076615; PMC6159302](#)). It is a genuinely simple, decades-old, single-number clinical index from a field with no overlap with vascular imaging, and it is structurally different from every metric already in the pipeline, since it counts discrete coil cycles per unit length rather than any curvature, angle-sum, or arc-chord ratio. It is not recommended as a routine standalone scalar for coronary vessels, however, because it presumes discrete, complete 360-degree coils along the vessel’s course,

a pattern coronary arteries only rarely exhibit; its useful conceptual niche is better filled by the Weibel/Paulsen/Fields carotid-artery classification scheme, which grades a vessel by its single sharpest local deflection angle rather than a continuous ratio: no abnormality below 15 degrees of deviation from straight, “tortuosity” for an inter-segment angle of 90 to 165 degrees, “kinking” for an angle under 90 degrees (with sub-grades at 60 to 90, 30 to 60, and under 30 degrees), and “coiling” reserved specifically for a full 360-degree loop ([Pathological Tortuosity of Carotid Arteries](#); corroborated by a Metz classification figure and a PubMed review, though the primary Weibel/Paulsen/Fields papers themselves were not directly fetched in this pass, so this rests on secondary clinical-review corroboration rather than primary verification). This scheme is recommended for addition because it is a genuinely distinct axis from the continuous sinuosity bands above (angle-based rather than ratio-based) and because CAS-Net’s own two independent output channels make a kinking/coiling category clinically meaningful as a qualitatively different entity from smooth bending, with the umbilical index’s coiling concept absorbed into this scheme’s “coiling” bin rather than tracked as its own metric. The related CASS iliac tortuosity severity bands (under 1.2 mild, 1.2 to 1.29 moderate, 1.3 to 1.49 severe, 1.5 or above very severe) are not recommended as a separate addition, since they classify the same underlying arc-chord ratio as the sinuosity bands above, only with a different vascular-surgery-specific set of numeric cut points, a duplicate under a different threshold set rather than new information.

The third group comes from movement ecology, a field that studies animal-trajectory straightness with entirely different vocabulary but partly overlapping mathematics. The plain Straightness Index, D/L , net displacement over path length, is mathematically the reciprocal of this project’s own $DM = L/D$ and is not recommended as a separate addition for exactly that reason, the same redundancy that already excludes Relative Length above. Benhamou’s corrected Sinuosity Index, by contrast, is a genuinely new closed-form combination not equivalent to anything already in the pipeline:

$$S = 2 \left[p \left(\frac{1+c}{1-c} + b^2 \right) \right]^{-1/2},$$

where p is the mean step length between consecutive centerline points, c is the mean cosine of the turning angle between consecutive steps, and b is the coefficient of variation of the step length ([Benhamou 2004, How to reliably estimate the tortuosity of an animal’s path](#), fetched directly). It is recommended for addition because coronary centerlines, once resampled at a fixed chord, still carry irregular local step lengths at branch points and endpoints, and this formula is specifically designed to remain valid under non-constant step length, a robustness property the project’s existing κ_a and TI do not explicitly carry. Root-mean-square curvature, $\kappa_{\text{rms}} = \sqrt{\text{mean}(\kappa_i^2)}$, is a second-moment summary of the same per-point curvature sequence already used for the peak/percentile addition above, distinct from both the mean (κ_a) and the order statistics (peak, 95th percentile), and is recommended alongside them as a third, complementary lens on the same underlying sequence, drawing on a comparative wall-shear-stress study that found curvature-based descriptors, including RMS curvature, consistently outperforming arc-chord tortuosity indices at explaining shear-stress variation ([Accuracy of vascular tortuosity measures using computational modelling](#), PMC8764056). A closely related “average squared-derivative curvature” named in the same source is not recommended for addition yet, because its exact formula, specifically whether it differentiates curvature along arc length before squaring or simply squares curvature itself, was never confirmed against the paper’s methods section in this research pass, and this report will not add a metric whose formula it cannot state with confidence.

Several further candidates surfaced in the sweep are explicitly left out because they duplicate

metrics already excluded from the pipeline for other reasons. Grisan’s Tortuosity Density, which subsegments a vessel at curvature sign changes and sums each subsegment’s own arc-chord ratio, is the same construct as the project’s already-excluded turn-segmented tortuosity index under a different name. Inverse-radius tortuosity, the mean of the inverse local radius of curvature, is conceptually equivalent to mean curvature computed through a different discretization and duplicates κ_a in spirit. Angle-tortuosity, distance-tortuosity, and a proprietary Vessel Curvature Index are not recommended for addition at this time regardless of their apparent novelty, because their formulas in the underlying notes come from a paraphrased description of a fetched review paper rather than a verbatim quotation, and this report treats that as insufficiently verified to commit to specific symbols and subscripts; they are named here as candidates for a future, dedicated full-text read, not as items ready to implement. Finally, no standalone, separately-named “coefficient of variation of curvature” or “standard deviation of turning angles” metric was found anywhere in this research pass despite a direct search for one, which this report treats as a genuine absence in the literature rather than a retrieval failure, and accordingly does not invent one to fill that gap.

The table below is the final, definite answer to what gets added and what does not.

Metric	Formula	Verdict
Inflection Count Metric (ICM)	$ICM = (n_{\text{inflections}} + 1) \cdot L/D$	Add (proposed earlier in conversation)
Peak / 95th-percentile local curvature	$\kappa_{\text{peak}} = \max_i \kappa_i(\ell)$, or 95th percentile of $\{\kappa_i\}$	Add (proposed earlier in conversation)
Horton-Strahler bifurcation ratio	$R_B \approx N_i/N_{i+1}$ by Strahler order	Add, per subtree (proposed earlier in conversation)
Sinuosity classification bands	thresholds $\approx 1.05 / 1.25 / 1.5$ on DM/TI	Add (proposed earlier in conversation)
Raw chord D and raw arc length L	reported standalone, not only inside $TI = L/D - 1$	Add (proposed earlier in conversation)
Bullitt Triangular Index (TI_{local})	$\frac{1}{n} \sum_i (a_i + b_i)/c_i$	Add: genuinely local, distinct from global TI
Weibel/Paulsen/Fields angle classification	sharpest deflection angle, bins at 15/90/165 degrees plus kinking sub-grades	Add: distinct angle-based axis, absorbs the umbilical coiling concept
Benhamou’s corrected Sinuosity Index	$S = 2[p((1+c)/(1-c)+b^2)]^{-1/2}$	Add: robust to irregular step length
RMS curvature	$\kappa_{\text{rms}} = \sqrt{\text{mean}(\kappa_i^2)}$	Add: second-moment complement to mean and peak
Relative Length (RL)	$RL = l/l_c$	Skip: reciprocal of $1 + TI$, redundant
Product-of-Angle-Distance (PAD)	$PAD = SOAM \times RL$	Skip: composite of non-comparable pieces
Straightness Index	D/L	Skip: reciprocal of DM , redundant
Umbilical Coiling Index	coils per unit length	Skip as standalone: niche absorbed by Weibel/Paulsen “coiling” bin
CASS iliac tortuosity bands	thresholds 1.2/1.3/1.5	Skip: duplicate of sinuosity bands, different cut points

Metric	Formula	Verdict
Grisan Tortuosity Density	turn-segmented arc-chord sum	Skip: already excluded under another name
Inverse-radius tortuosity	mean of $1/R_{\text{local}}$	Skip: duplicates κ_a
Angle-/distance-tortuosity, VCI	paraphrased, unverified	Defer: needs primary-source verification first
Average squared-derivative curvature	formula ambiguous	Defer: cannot state the formula with confidence
CV of curvature / SD of turning angles	no established formula found	Not added: genuine literature gap, not fabricated

What this changes about the pipeline as a whole

The overall shift this report locks in is narrower than “stratify everything by vessel and by position,” and that precision matters. Stage 1’s local descriptors were already vessel-specific and needed no change; the actual defect was at the aggregation and reference-range stages, where a single pooled whole-tree number silently averaged over exactly the kind of vessel-identity heterogeneity the coronary rank-reversal and retinal GWAS evidence both document. The positional question resolves the same way: not a call to abandon the whole-vessel scalar, but to add a continuous, arc-length-normalized profile beside it, precisely because the compositionality evidence forecloses the tempting shortcut of averaging landmark segments into a corrected replacement for that scalar. And the ten metrics added here are deliberately not a shapeless “more features” expansion; each earns its place by measuring something the existing κ_a , τ_a , TI , and D_κ battery cannot, an order statistic, a second moment, a discrete-angle category, a branching-order ratio, or a step-length-robust combination, while roughly as many candidates were turned away for being the same ratio under a different field’s name. The open gaps named along the way, diagonal and left-main tortuosity norms, a disease-independent proximal-distal comparison, and a quantified bifurcation-window bias curve, are not filled by this report and should not be treated as filled; they are the concrete next things to measure on this project’s own coronary centerlines before any of the added structure above is trusted the way the retinal WAL/WAC formula was trusted only after its own compositionality and expert-correlation checks were run.